

ECEN 215, Spring 2022

Electrical Engineering Principles

Lecture 2: Voltage and Current

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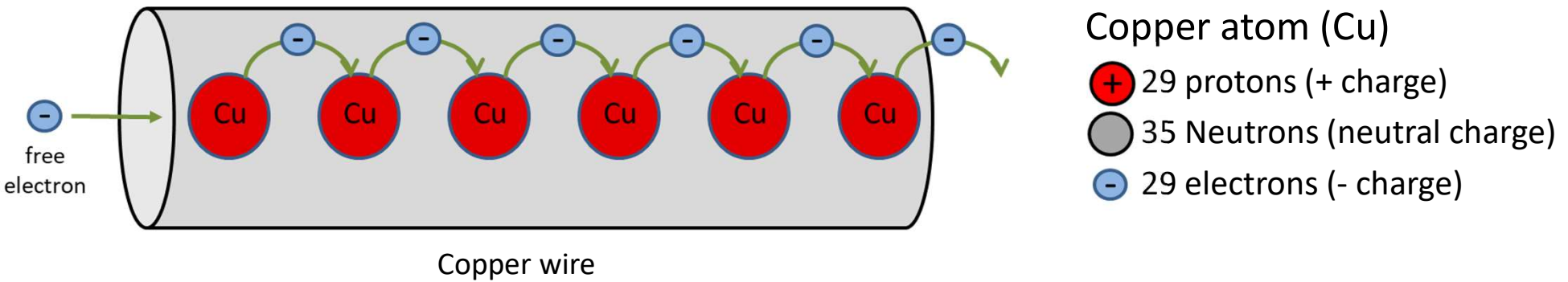
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OUTLINE



- Circuit Elements
 - Current and Voltage
 - Energy and Power
-

What is Electricity?



Separation of charge creates an electric force: Voltage (Volts)

Motion of charge creates an electric current: Current (Amps)

Definition of Current

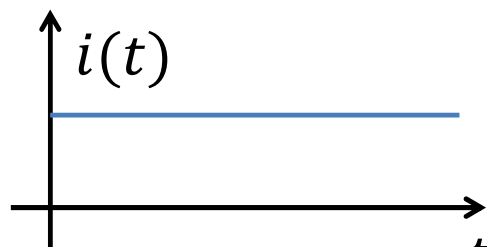
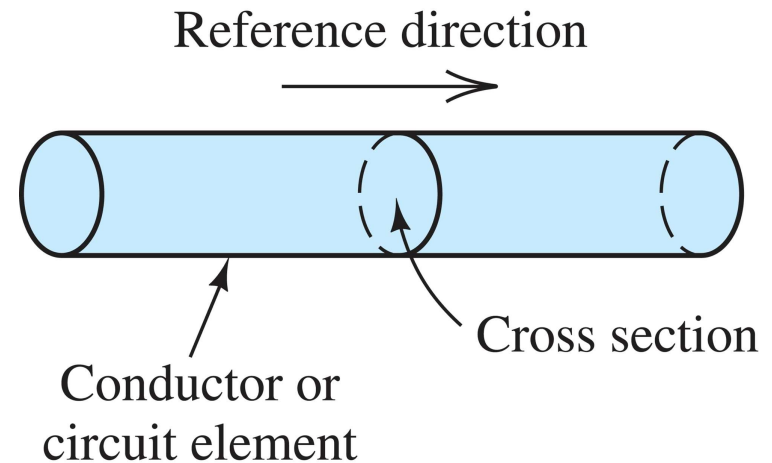
- Definition: Electrical current is the rate of flow of charge.

$$I = \frac{dq}{dt}$$

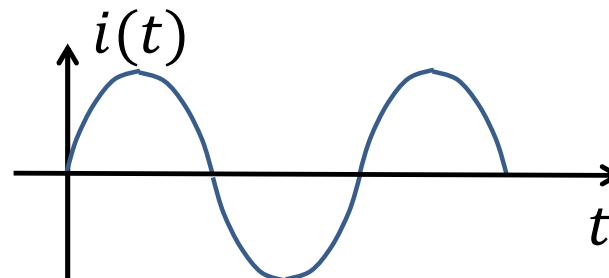
I =current (Amps)

t =time (seconds)

q =charge (Coulombs)



Direct current (DC)



Alternating current (AC)

Don't forget – positive current is in the direction opposite of the physical flow of electrons.



Definition of Voltage

- Definition:** Voltage is the energy (work) required to move a unit of charge.

$$V = \frac{dw}{dq}$$

V=voltage (volts)

w=work or energy (Joules)

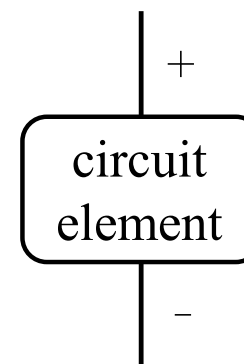
q=charge (Coulombs)

In terms of electrical circuits, we will be concerned with the amount of energy required to move a unit of charge from one terminal of a circuit element to another.

Units of voltage

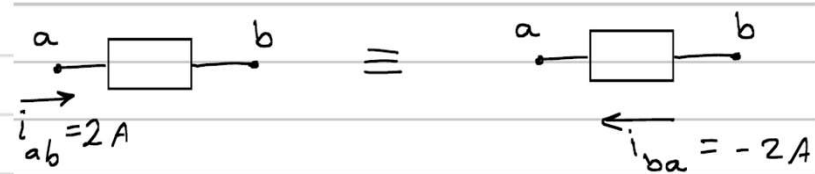
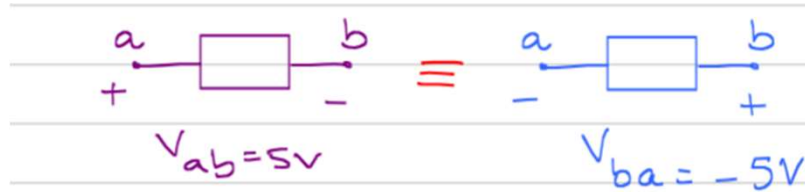
$$1 \text{ Volt} = \frac{1 \text{ Joule}}{1 \text{ Coulomb}}$$

So in a 9V battery, each Coulomb of charge carries 9 Joules of potential energy.



Voltage and Current

- Note representation and sign



Ex. The current flowing through an element is

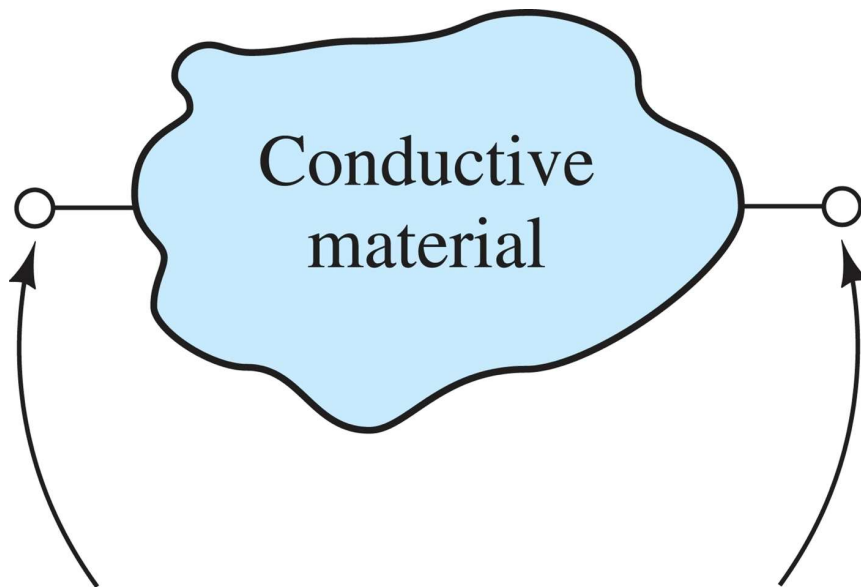
$$i = \begin{cases} 2A, & 0 < t < 1 \\ 2t^2 A, & t \geq 1 \end{cases}$$

Calculate the charge entering the element from $t=0$ to $t=2$ s.

$$\begin{aligned} i &= \frac{dq}{dt} \rightarrow q = \int i dt \\ q &= \int_0^2 i dt = \int_0^1 2 dt + \int_1^2 2t^2 dt \\ &= 2t \Big|_0^1 + 2 \cdot \frac{1}{3} t^3 \Big|_1^2 = 2(1-0) + \frac{2}{3}(8-1) = \underline{6.67 C} \end{aligned}$$

- Work through the other example in the notes

Resistor



Terminals for current
to enter or leave

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Table 1.3 Resistivity Values (Ωm) for Selected Materials at 300 K

Conductors	
Aluminum	2.73×10^{-8}
Carbon (amorphous)	3.5×10^{-5}
Copper	1.72×10^{-8}
Gold	2.27×10^{-8}
Nichrome	1.12×10^{-6}
Silver	1.63×10^{-8}
Tungsten	5.44×10^{-8}
Semiconductors	
Silicon (device grade) depends on impurity concentration	10^{-5} to 1
Insulators	
Fused quartz	$> 10^{21}$
Glass (typical)	1×10^{12}
Teflon	1×10^{19}

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$$R = \frac{\rho L}{A}$$

ρ = resistivity
 L = length
 A = cross sectional area

Resistance depends on the resistivity that is associated with any material

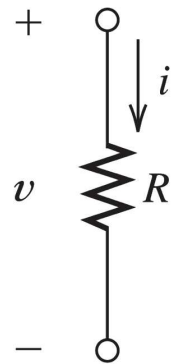
Resistor and Ohm's law



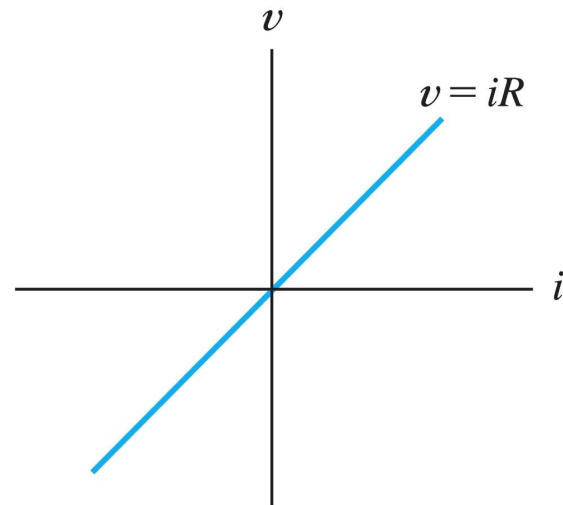
Resistor

Resistance is the capacity of material to resist the flow of electric current. It is denoted by R and measured in ohms.

Resistor is a circuit element used to model the current resistivity of the resistor.



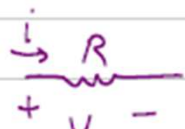
(a) Resistance symbol



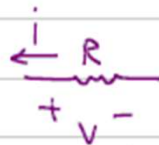
(b) Ohm's law

Resistor and Conductance

implementing passive sign convention:



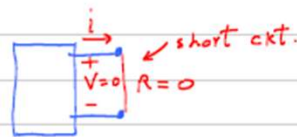
$$V = Ri$$



$$V = -Ri$$

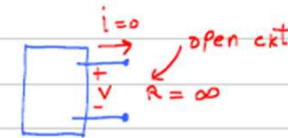
R is always positive $0 < R < \infty$

$R = 0$ (short ckt.)



current is determined by
the ckt. inside the box

$R = \infty$ (open ckt.)



voltage is determined by
the ckt. inside the box

Conductance is the reciprocal of resistance, denoted by G and measured in Siemens (S).

$$G = \frac{1}{R}$$

$$i = GV \text{ (ohm's law)}$$

Power of Resistor



Power for resistors

$$P = Vi = Ri \cdot i = Ri^2$$

$$P = Vi = V \cdot \frac{V}{R} = \frac{V^2}{R}$$

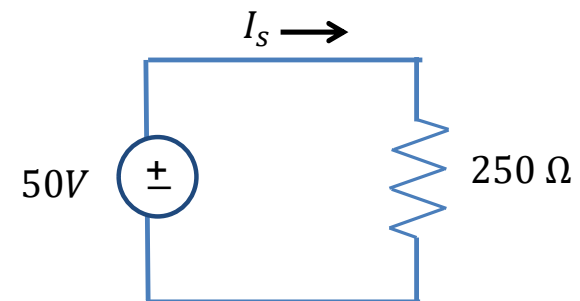
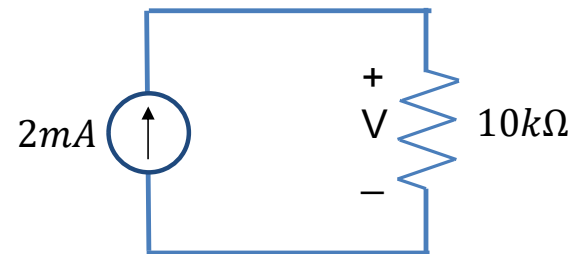
$$P = Vi = i^2 R = \frac{V^2}{R}$$

The power of a resistor is always positive and it can only absorb power.

$$\text{OR, } P = \frac{i^2}{G} = V^2 G \quad (\text{in terms of conductance})$$

Examples #1 and #2

- The current source is producing 2 mA of current, which then flows across the upper wire, down through the resistor, and back leftward across the lower wire
- The voltage source ensures that the difference between the upper and lower nodes (wires) is 50 V
- Independent sources are drawn with a circle. They produce a constant current (arrow) or voltage (plus and minus signs).
- Resistors have a linear relationship between its current and voltage, $V = IR$ (known as Ohm's law)
 - First example: $V = (2 \text{ mA}) (10 \text{ k}\Omega) = 20 \text{ V}$
 - Second example: $I = (50 \text{ V}) / (250 \text{ }\Omega) = 0.2 \text{ A}$

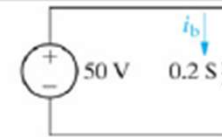


Example: Power of Resistor

Ex. Calculate the unknown voltage or current and determine the power dissipated in each resistor.



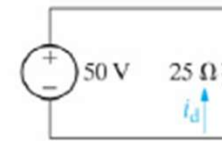
(a)



(b)



(c)



(d)

$$a) v_a = 8 \times 1 = 8 \text{ V}, \quad P = v_i = 8 \times 1 = 8 \text{ W}$$

$$b) i_b = VG = (50)(0.2) = 10 \text{ A}$$

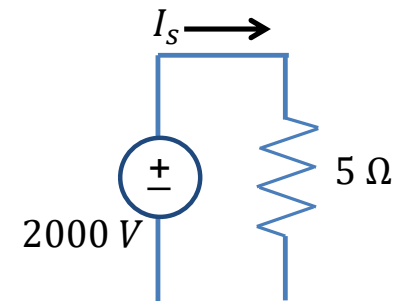
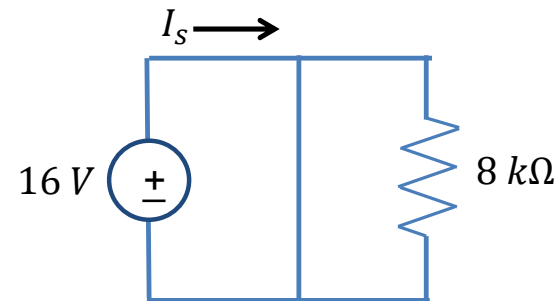
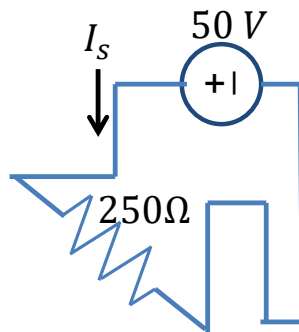
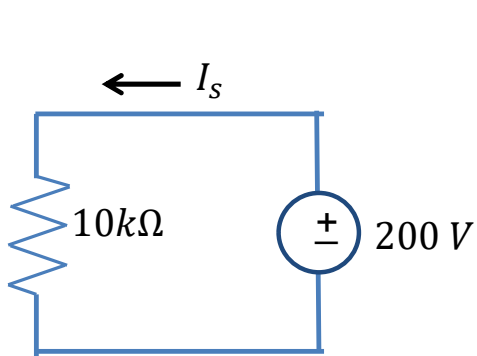
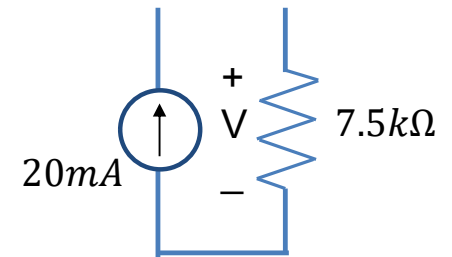
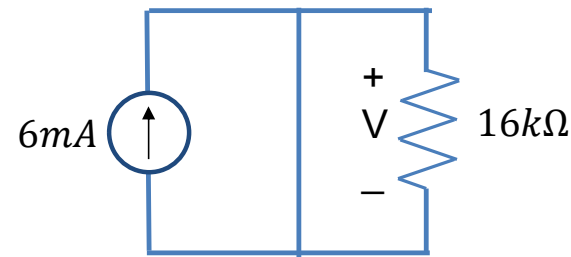
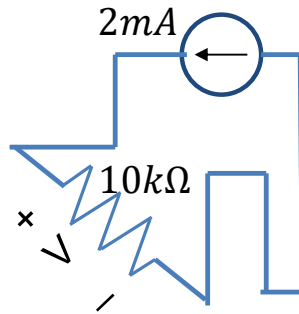
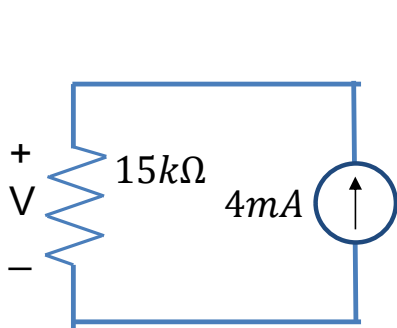
$$P = V^2 G = (50)^2 (0.2) = 500 \text{ W}$$

$$(\text{or } P = Vi = 50 \times 10 = 500 \text{ W})$$

$$c) v_c = (-1)(20) = -20 \text{ V}, \quad P_{20\Omega} = Ri^2 = (20)(-1)^2 = 20 \text{ W}$$

$$d) i_d = -\frac{50}{25} = -2 \text{ A}, \quad P_{25\Omega} = Ri^2 = 25(-2)^2 = 100 \text{ W}$$

Try a Few Variations (Remember $V=IR$)



Circuit Elements

Ideal circuit element:

1. It has only two terminals .
2. It is described mathematically in terms of its voltage and current.
3. It cannot be subdivided into other circuit elements.

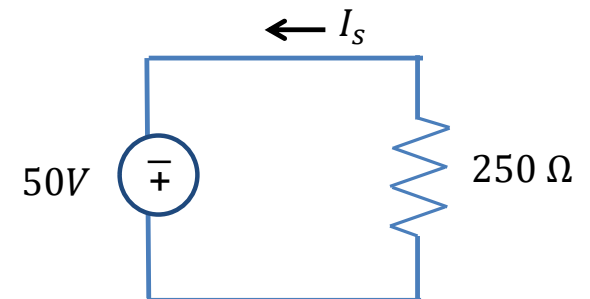
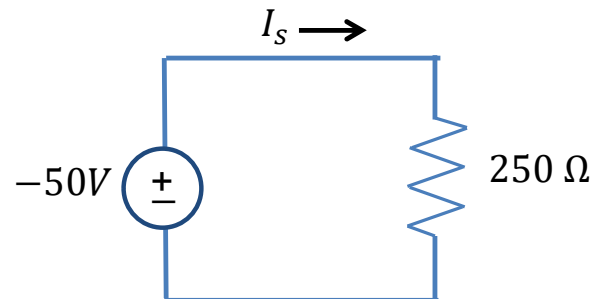
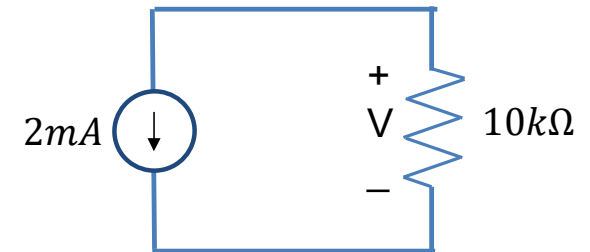
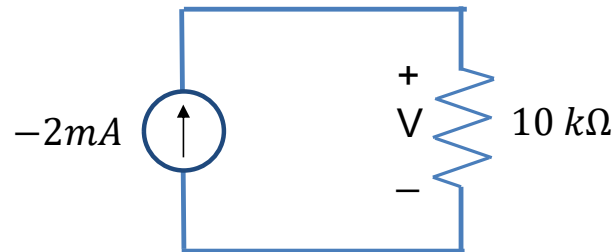
Passive sign convention (or, Passive Reference Configuration): when the current enters through the positive terminal of an element, passive sign convention is satisfied. In this case, use a positive sign in any expression that relates the voltage to the current. Otherwise, use a negative sign.



PSC is satisfied

Watch Your Signs!!

- Voltages and currents can be positive or negative
 - Negative current means the current is flowing in the opposite direction of the arrow
 - Negative voltage means the minus side is higher potential than the plus side
- Ohm's law applies if current is oriented *entering* the resistor where the plus side of the voltage measurement is taken
- Resistances will rarely be negative (this would be a power-producing device)



Don't forget your units as well!

Electrical Power



- Definition: Power is the rate of energy (work) expended per unit time.

$$P = \frac{dw}{dt}$$

P=power (Watts)

t=time (seconds)

w=work or energy (Joules)

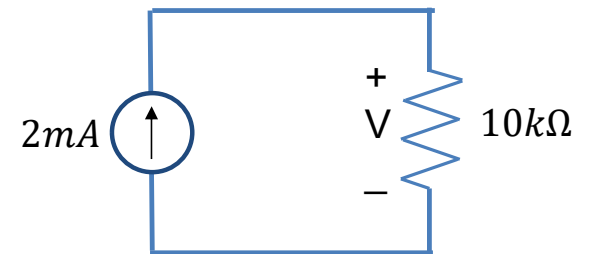
$$1 \text{ Watt} = \frac{1 \text{ Joule}}{1 \text{ second}}$$

Calculate the power produced by the source and absorbed by the resistor.
(40 mW)

- Note also that

$$P = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = v \cdot i$$

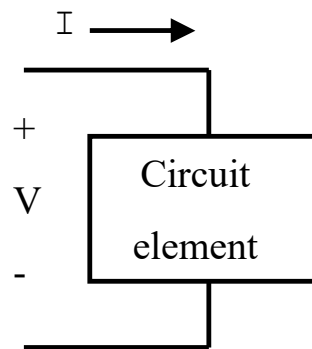
- 1 Watt = (1 volt)(1 amp)



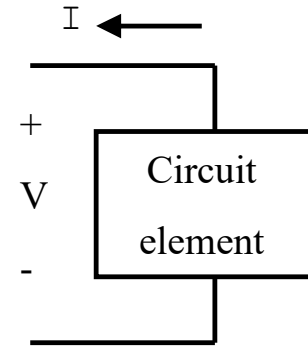
Sign convention really matters with power! If current flows **into** the terminal with **the plus side** of the voltage measurement (like in Ohm's law) the device is **absorbing** power.

Passive Sign Convention

- Passive Sign Convention - For a two terminal element, whenever the reference direction for the current is in the direction of the voltage drop, use a positive sign in any expression that relates voltage to the current. Otherwise use a negative sign.
- In other words, the current should be going *into* the plus side of the voltage measurement



$$\begin{aligned}
 V &= +IR \\
 V &= +L \frac{dI}{dt} \\
 I &= +C \frac{dV}{dt} \\
 P &= +IV
 \end{aligned}$$

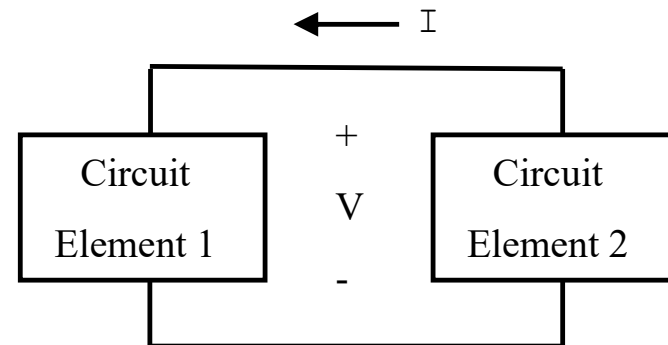


$$\begin{aligned}
 V &= -IR \\
 V &= -L \frac{dI}{dt} \\
 I &= -C \frac{dV}{dt} \\
 P &= -IV
 \end{aligned}$$

Test Your Understanding

- In this example two circuit elements are connected together. With the voltage polarity and the direction of current defined in the drawing, for each case below determine if Element 1 is delivering power to element 2 or vice versa.

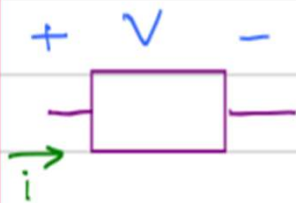
- a) $V = 3 \text{ volts}$, $I = -20 \text{ mA}$,
- b) $V = -3 \text{ volts}$, $I = 20 \text{ mA}$,
- c) $V = -3 \text{ volts}$, $I = -20 \text{ mA}$,



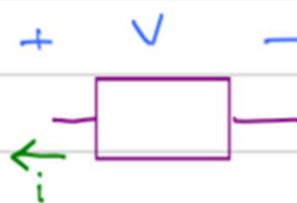
Power and PSC



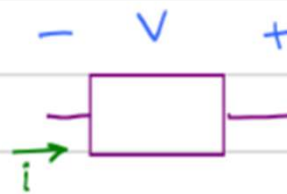
Applying P.S.C



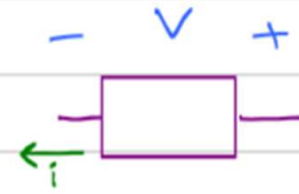
$$P = vi$$



$$P = -vi$$



$$P = -vi$$



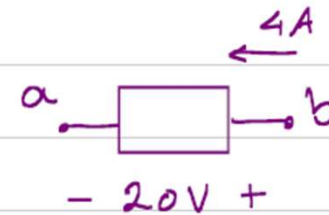
$$P = vi$$

If $P > 0$, the circuit element absorbs power and if $p < 0$, the circuit element delivers power.

Power and PSC (2)

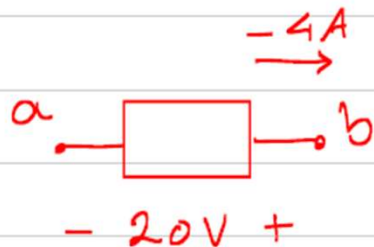


Ex. Assuming a 20 V drop from terminal b to a, and a current of 4 A enters terminal b.

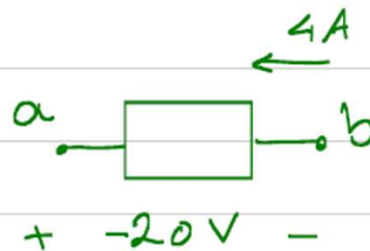


Find the value of v , i and p for different polarity references.

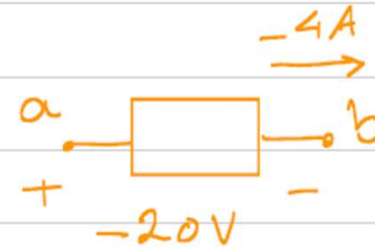
$$P = vi = 20 \times 4 = 80 \text{ W (abs.)}$$



$$P = -vi = -(20)(-4) = 80 \text{ W (abs.)}$$



$$P = -vi = -(-20)(4) = 80 \text{ W (abs.)}$$



$$P = vi = (-20)(-4) = 80 \text{ W (abs.)}$$

Conservation of Energy



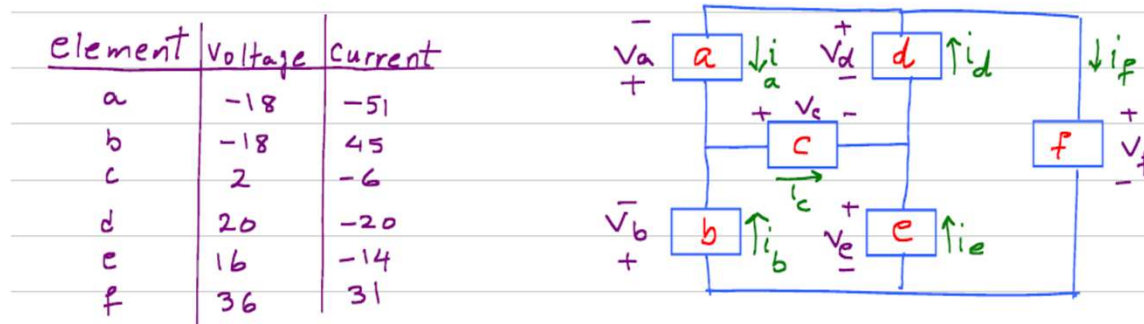
Conservation of Energy or Power Balance

- Total power generated = total power absorbed
 - Circuit must be isolated from any other devices
 - Net sum of the power of all elements = 0
- Use it to check your work!



Example1: Calculate Power

Ex. The numerical values of voltages and currents are given in the table. Find the power of each element.



$$P_a = -V_i = -(-18)(-51) = -918 \text{ W (del.)}$$

$$P_b = V_i = (-18)(45) = -810 \text{ W (del.)}$$

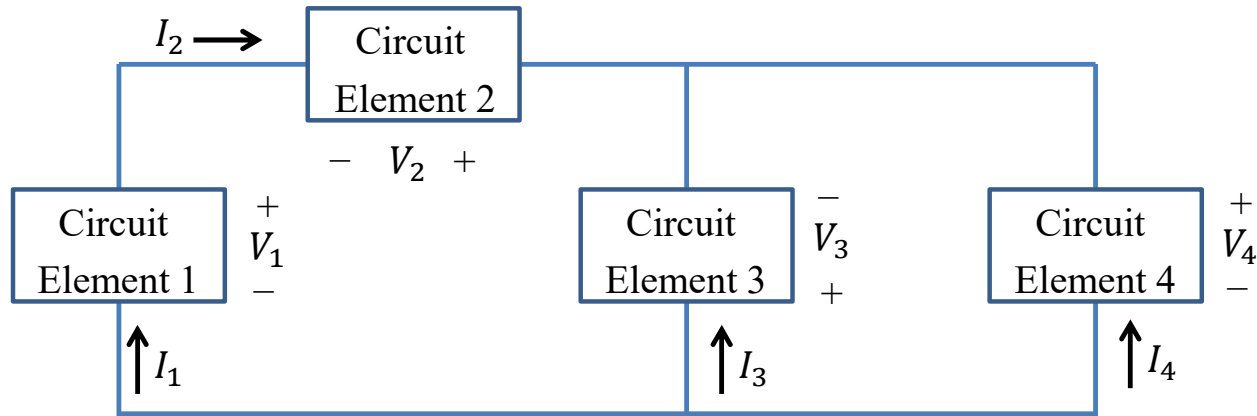
$$P_c = V_i = (2)(-6) = -12 \text{ W (del.)}$$

$$P_d = 400 \text{ W (abs.)}, P_e = 224 \text{ W (abs.)}, P_f = 1116 \text{ W (abs.)}$$

$$\text{Power balance check: } \sum P \stackrel{?}{=} 0$$

$$-918 - 810 - 12 + 400 + 224 + 1116 = 0 \quad \checkmark$$

Example 2: Calculate Power



Element	Voltage	Current
1	3 volts	-1mA
2	-2 volts	-1mA
3	-1 volt	4mA
4	1 volt	-3mA



Element	Power
1	3
2	-2
3	-4
4	3

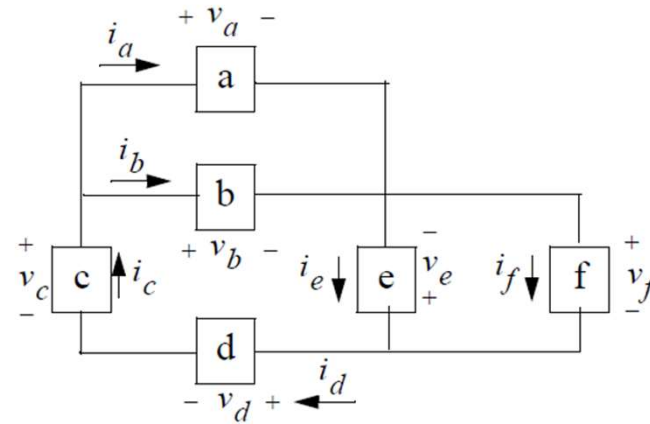
Total Power = 0

Example3: Calculate Power



The numerical values for the voltages and currents in the circuit shown are given in the table below. Find the total power developed in the circuit.

Element	Voltage (kV)	Current (mA)	Power (W)
a	150	0.6	$+150 \cdot 0.6$
b	150	-1.4	$-150 \cdot 1.4$
c	100	-0.8	$+100 \cdot 0.8$
d	250	-0.8	$-0.8 \cdot 250$
e	300	-2.0	$+600$
f	-300	1.2	$-300 \cdot 1.2$



Discussion and Q&A



- Questions?

Next Lecture



- Sources
 - Circuit Connections
 - Kirchhoff's Laws
-